

Personalized Interactive Gesture Recognition Assistive Technology

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ABSTRACT

Computing systems have the potential to contribute with interactive and low-cost solutions to support Augmentative and Alternative Communication (AAC) by applying different technologies to address different users' characteristics and needs. Computer Vision-based AAC systems can support users with motor difficulties by tracking and recognizing their remaining functional motions. To investigate possibilities for people with motor and speech impairments, this paper presents the PGCA: a Computer Vision system that allows the creation of a personalized gestural interaction as assistive technology for communication purposes. PGCA system takes into account the motor abilities and limitations of its users and the knowledge of caregivers in recognizing the gestures performed by the users. Results from interviews with special education professionals and from an experiment with the target audience suggest the use of personalized gestures is a common practice for AAC, and that creating custom datasets can be challenging, mainly due to the level of understanding of participants, the similarity between gestures, and variations in performing the same gestures. Improvements for the system were identified and described aiming to make the interface easier and more effective.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → **People with disabilities**; • **Computing methodologies** → **Motion capture**.

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KEYWORDS

Assistive Technology, Augmentative and Alternative Communication, Computer Vision, Gesture Recognition, Accessibility

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1 INTRODUCTION

Augmentative and Alternative Communication (AAC) can complement or replace speech using non-verbal communication systems and intervention strategies [24]. AAC systems and devices provide tools that allow users to select words, symbols, and images to communicate their thoughts, intentions, and conversations with others through digitized or synthesized voice [2]. People with speech impairments often have motor impairment associated, which makes the use of traditional input devices such as mouse, keyboard or touch difficult, since these users may not have sufficient mobility or motor control to manipulate these devices [10]. Therefore, the development of alternatives technologies is necessary to address these issues. The existence of different technologies and interaction modes opens new paradigms in Human-Computer Interaction (HCI), that can benefit people with motor and speech impairments.

Easy access to camera devices gave rise to a new generation of Assistive Technology (AT) resources that allows personalized and non-invasive interaction methods to be considered, including movements recognition of the hands, head, face, and other body parts to perform actions on computer systems [36] [33]. Such possibilities represent a valuable opportunity for disabled people because they promote the more inclusive and active participation of disabled people in society by influencing their affective and emotional states and strengthening their ties with the environment in which they are inserted [4]. In Brazil, different initiatives have been proposed to promote innovation in AT, whether through

families [41], companies [57], universities [8] [7], or federal government [22] [23].

Regarding gesture recognition, however, most existing studies work with predefined gesture sets and tracking a specific body region. After leaving the laboratory conditions, in case there is a need to replace any pre-configured gesture or to include new ones, no studies were identified describing how it would be possible for the user and his caregiver to perform this type of customization. To contribute in this direction, this paper presents a Computer Vision system that allows users and caregivers to create a personalized set of interactive gestures, taking into account the abilities and limitations of the user when performing movements, and providing support to other people to recognize these movements using an AAC system based on personalized gestural interaction. The developed system is presented along with the results obtained in an experiment carried out with students with motor and speech impairment. The information obtained through interviews with teachers, caregivers and LIBRAS (Brazilian sign language) interpreters that accompanied the experiment are also presented and they have contributed to the identification of some challenges and possibilities to improve the system.

2 RELATED WORK

Possibly, the earliest example using video to recognize hand movements as an interaction mode is the study of Krueger et al. [31]. Jacob [28] was one of the first to introduce appearance-based interaction techniques into real-time applications for people with disabilities. Since then, several studies supported by resources of AT and interaction modalities based on recognition of body movements have been developed to improve the quality of life of people with physical disabilities.

Betke et al. [9] developed the CameraMouse, a system capable of tracking the movement of various parts of the body specified by the user (such as nose, eyes, chin, foot) employing an algorithm based on spatiotemporal model matching. This study was quite representative for the AT area and inspired other studies such as [32], [19], [3]. Similar systems were also developed by [42] and [21]. Simple devices such as webcam were used by a large number of initiatives ([3] [34] [16]) mainly because they represent an interesting and viable alternative for detecting and tracking of movements, and especially due to their low cost. The same advantage holds for mobile devices, that are increasingly accessible and used in AT resources [38] [39] [63]. Recent studies have investigated the use of depth data ([11], [25] [40]), using devices such as Kinect, BumbleBee and Monocular infrared.

Gaze detecting/tracking research has received increasing attention from the academic community ([6] [12] [18]), possibly because eye movements may be the only movements some people with severe disabilities can control voluntarily. In studies of [13] and [48], the gaze is used in conjunction with movements performed by other parts of the body or other devices, generating modalities of multimodal interaction. The existing studies generally do not provide details on the difficulties encountered in creating the dataset, possibly because they use existing datasets, or created with gestures performed by people without disabilities. Tracking specific regions of the human body as a way of interaction for a specific target audience tends to generate solutions more adapted to the users' interests. However, the accessibility of these solutions may fail if they do not offer adaption mechanisms that allow themselves to be tailored for their users.

Although not employing cameras but sensors on users (magnetic tracker and electromyography), studies from Roy et al. [46, 47] show that many non-speaking individuals are able to make gestures that are repeatable and recognizable by family observers in greater number than previously expected, and these gestures can be mapped into words or concepts. When a person has severe limitations to self-expression, the knowledge that the observers (e.g. caregivers, family members) have about an individual's ability to perform movements is fundamental to create a personalized sign language. The research we present in this paper aims to support in this direction, by allowing a personalized gesture configuration so that people can interact with a computer system.

3 PERSONAL GESTURE COMMUNICATION ASSISTANT

The Personal Gesture Communication Assistant (PGCA) is a Computer Vision system developed to support AAC as an assistive technology for personalized gestural interaction, exploring alternatives that are adaptable and inexpensive. The system's first version was evaluated by HCI experts and the results (presented in [5]) informed the system redesign for its the new version presented and evaluated with representatives from the target audience. This system is based on a methodology to support AAC [5] in which gestures and their meanings are created and configured by users and their caregivers. The PGCA system captures movements from a webcam and represents them as Optical Flow-based Motion History Image (OF-MHI). Two classifiers recognize these movements (gestures): one based on Support Vector Machine (SVM-based classifier), and other based on Convolutional Neural Network (CNN-based classifier). For the implementation of the system, it was mainly used the programming language C++ and QT platform. CNN-based classifier was implemented using programming language Python embedded on the C++ application.

Personalized gestural interaction

When initiated the system, the camera automatically allows the user to calibrate, adjusting the initial positioning using

the user's nose as reference to centralize the face for the camera. After the calibration process, the caregiver can start creating a custom dataset. For the acquisition of each example that will compose the dataset, the movement is captured and represented as an image. After capturing several samples of each gesture (at least ten), the system can be trained by one of the implemented classifiers and is ready to test the interface on gesture recognition for the learned gestures. The techniques used in the system for gesture recognition and motion representation are briefly described in the sequence.

Motion representation by OF-MHI. Proposed originally by Davis and Bobick [14] [17], MHI is a global spatio-temporal representation of motion that has been applied to motion analysis and tracking for different purposes, such as gesture recognition [59] or human action recognition [27] [62]. MHI converts the 3D space-time information from a video sequence into a single 2D intensity image, including information of time and space and reflecting also the movement order. The optical flow was used to aggregate velocity information to MHI. Optical flow [26] [35] denotes a shift of the same scene in image sequence at a different moment in time, estimating pixel-level movement between two images. In conventional MHI, every detected foreground pixel is assigned with a fixed intensity value, a slow movement and a fast movement of different body parts will have the same motion strength [58] (i.e., speed differences are not considered). Different proposals have been presented to add velocity information on MHI using optical flow, as Tsai et al. [58], Fan and Tjahjadi, [20] and Khalifa et al. [30], and a similar proposal was used in our study. In our system, a labeling algorithm is applied to a silhouette obtained by frames difference for identifying connected regions. Afterward, the Lucas Kanade optical flow [35] is calculated for the centroid pixels of each of these regions, and this displacement value is replicated to the other pixels of the same region to make it faster the tracking process via optical flow. When the "Zoom in" option is checked in the system settings, the facial landmarks are entered as points to be tracking, allowing to highlight and improve the perception of facial movements. Figures 1 show the points used to track movement and calculate optical flow.

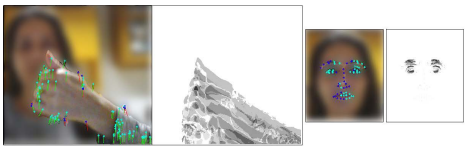


Figure 1: Points used to tracking and calculate optical flow of a hand gesture and a facial gesture.

SVM-based classifier. For the system's first version, it was used a multiclass SVM discriminant classifier and a feature descriptor Histogram of Oriented Gradient (HOG) for gesture recognition. Discriminant classifiers are trained to separate data into two classes by learning a hyperplane in a larger dimensional space. The "one-versus-all" method was used to address problems of multiple classes: given an n-class problem, for each class a binary model is constructed, which the training set consists of examples of this class as positive labels and examples of other classes as negative labels. HOG is a feature descriptor used for object detection, and it was obtained based on image gradient histogram. For this classifier, the HOG was extracted from OF-MHI.

CNN-based classifier. CNN descriptors are able to generate features more sophisticated and difficult to elaborate in descriptive means, but it is necessary a large and varied dataset to train a CNN from scratch. In our context, the number of samples is limited because each user will create his own dataset, then, Transfer Learning was considered as a viable alternative because it can improve the knowledge of the learner (i.e., the learning mechanism) about one domain by transferring information from a related domain [61]. Therefore, it was used the TensorFlow [1] Inception V3 [56], a codename for a deep CNN architecture model, originally trained on ImageNet dataset [49], as a base to retrain a custom set of images. Afterward, it was applied Transfer Learning by retraining Inception's final layer with new categories to build a custom image classifier according to labels and gestures generated by system users.

System interface

PGCA interface is composed of three main areas with specific purposes: 1. Caregiver area, where the dataset is created; 2. User area, where gesture recognition is used to interact with the system and with communication boards; and 3. Communication boards area, where new boards can be generated by selecting images.

Caregiver area. Figure 2 presents the Caregiver area, where datasets are created by using OF-MHI to motion representation. Gestures are related to classes labeled with words that will be used for communication or configuration purposes (e.g., Hi, Bye, Restroom, Food, Water, Confirm, Undo). After performing the experiment with the target audience described in this paper, the need to guarantee the quality of the samples captured by the system through an image matching algorithm was identified. Therefore, the first movement captured to represent a given gesture is played back to the user and it is requested a confirmation that the movement performed is correct. After the user confirmation, the first gesture will be saved as the base sample and it will be used as a reference to compare the next movements to be captured for that same gesture. To compare the captured samples, a subtraction operation is performed between corresponding

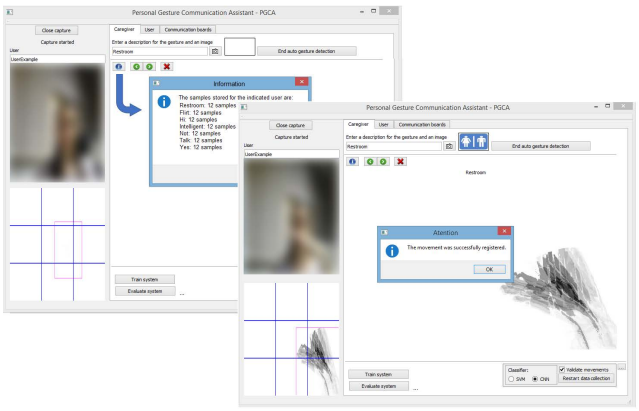


Figure 2: Caregiver area where datasets are created and the system is trained to gesture recognition.

OF-MHIs, and the remaining area is checked to see if it is not greater than the original area of the base sample. In addition, the centroid of each of the OF-MHIs is also compared to check if they are in the same quadrant in the image or in the close quadrant. Thus, whenever the system identifies a very different sample from the base sample, the user will be informed that the sample was considered invalid, and it will be presented an option to discard or maintain it. The system allows recording only samples considered valid for each gesture and to discard samples that have been erroneously generated during the realization of some involuntary movement by the user. A setting available on the caregivers' interface allows the user to choose whether to use or not this sample validation feature. After completing the dataset generation, the caregiver can start the system training, which will automatically perform the data set expansion processes (by means of Data Augmentation) and train the system to recognize the gestures using one of the implemented classifiers. Following, the caregiver can test the system to evaluate the accuracy of the classifier on gesture recognition. If this test of accuracy produces acceptable results (ideally greater than 90%), the model can be used for gesture recognition. If the test of accuracy indicates poor performance of the system, a description of the captured gestures is presented to the user, containing hit rate for each one. This information will allow the user to perform a new data collection or to exclude of the dataset the gestures that are not being correctly recognized.

User area. The User area is available for use after the system training. This area represents the main interface with AAC functions in which gesture interaction will allow the execution of the following functions: Settings, Detection and representation of gestures, Navigation on communication boards, Selection of Images, and Simulation of keyboard use. Figure 3 (i) shows an example of user area usage where the execution of a gesture is recognized by the system. The corresponding figure is presented to the user, the class name (Label) is inserted into a text box and the label spoken by a voice synthesizer. Figure 3 (ii) shows an example of navigation in images registered in the communication board (composed of images available on the ARASAAC portal [45]) that can be created by the caregiver. The settings screen is hidden by default, and it is displayed only when the user access the option represented by a "gear" icon. In this screen, presented in Figure 3 (iii), the user can configure the system to relate gestures to navigation functions used to view previous, next, select, undo, key simulation, among others. The basic configuration process may be done with the caregiver support, and once the user knows the basic gestures related to the actions defined in the settings, the user may access the settings interface and customize the system by using gestures. When the user relate a gesture to certain system functionality, the movement for the selected gesture is reproduced on the side of the selection box. Below each selection box, a motion video can be played to show the gesture configured.

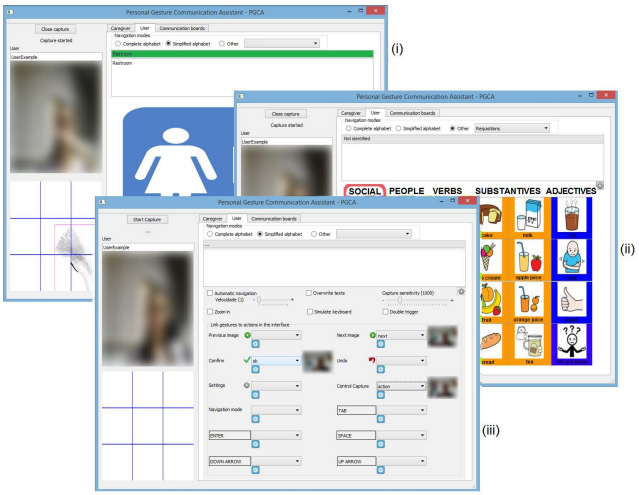


Figure 3: User area where gesture recognition is used for communication (i), interaction with Communication boards (ii), and customizing functionalities (iii)

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Communication Boards area. Figure 4 shows the Communication Board's area where the caregiver can create different communication boards composed of images. After the user can select images from these boards for communication purposes.

4 INTERVIEWS WITH DOMAIN EXPERTS

After evaluating the system interface in a laboratory with HCI experts and after evaluating different technologies to

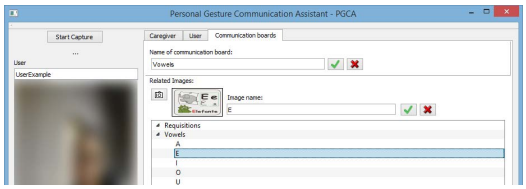


Figure 4: Communication boards area where caregiver can generate customized boards using images.

perform gesture recognition and motion representation, an interview with teachers from the special education domain was conducted.

According to the project approved by the Research Ethics Committee, four schools linked to co-participating institutions were visited before testing the system with people with motor and speech impairment. One of the schools is a specific educational institution for students with special needs (Parents and Friends Association of the Exceptional - APAE), and the others are inclusive schools where students with special needs can also attend regular classes: State Center for Basic Education for Youth and Adults (Public School A), State College - Elementary, Middle and Professional Education (Public School B), State College - Elementary and Middle Education (Public School C). In the four schools, interviews were made with regular teacher, support teacher, caregiver or LIBRAS interpreter to better understand the characteristics of the target audience. The interviews were conducted with eight professionals linked to special education - see Table 1 for details.

Table 1: Information of the professionals interviewed

Inte.	Sex*	Time experience	Link with students	Institution
P1	F	28 years in special education	Classroom teacher	APAE
P2	F	19 years in special education	Classroom teacher	APAE
P3	M	4 years in special education	Rotating teacher	APAE
P4	F	11 years in special education	Classroom teacher	APAE
P5	F	18 years as LIBRAS interpreter	LIBRAS interpreter	Public School A
P6	F	6 years as operational assistant	Caregiver	Public School A
P7	F	14 years in special education	Teacher of Support to Communication	Public School B
P8	M	7 years in special education	Teacher of Support to Communication	Public School C

*Sex: M - Male; F- Female.

When questioned about the main form of communication used with the students that present motor and speech impairments, all the eight professionals reported the use of gestural language. Teachers reported that some students are aware of some LIBRAS signs, but most students have their vocabulary of signs, that is, gestures used in the family environment that do not follow a specific sign language.

Teachers who work in the specific school for students with special needs reported they use a lot of association of concepts with images, suggesting using communication boards with figures to encourage students' interaction with the system is a valid approach. Besides, many students with special needs enrolled in this school are not yet fully literate, which prevents the use of alphabetical boards. Because the boards available in the system by default may not meet the demands of teachers with their students for all the possible different scenarios they may find, a feature was designed to support users to create their boards of communication.

The use of a specific AAC system was reported only by the two teachers who work to support the communication of students. A teacher mentioned she knows the Boardmaker system [37] but alleged not to use it in the school environment due to the associated cost. Another teacher cited the Livox application [41] despite having never used it with his student because the student can not interact with touch-based devices. The professional who works as a LIBRAS interpreter mentioned the use of some specific software for teaching and recognizing LIBRAS, such as the "Sinalário disciplinar de LIBRAS" [51] and the ICOM App [50] to make video calls with the deaf. She also uses book with CDs [60] to review concepts, meanings and represent movements.

None of the eight interviewed professionals have ever used software that uses a camera for gesture recognition. Some teachers said they used a camera just to record students' videos, record photos and present them later in presentation slides. The LIBRAS interpreter commented she had already used the Windows Live ^{TM} Messenger software with the camera enabled to communicate by LIBRAS signs with some deaf students. In the first conversation with one of the teachers, he was concerned whether something would be put on the student's body. According to the teacher, the mother of a student volunteering for this research was already tired of taking him to perform tests with wearable devices. In this case, the use of a camera exclusively was decisive for the family to agree to participate in the experiment.

Students with special needs enrolled in the visited schools visited are usually from poor families, and according to their teachers, simple inclusion in the school environment plays a very important role in their social development. As well as learning curriculum content, they also receive a lot of guidance related to the execution of daily activities in order to motivate these students to be more autonomous and functionally independent.

Although the focus of the research related to this study is not directed at the learning process of children with special needs, it is possible to observe that the developed system has the potential to serve as a support for teachers in the task of literacy. In this context, some teachers reported that a method of literacy that has shown good results in the literacy

of children with special needs is the method Challenges of Learning - "ABACADA" [52]. This method is aimed at students with learning disabilities and intellectual disabilities and is based on the Fonic Method [15] and Sodré Method [54]. The method ABACADA associates the initial syllable of each word with a corresponding figure (Figure 5 A), and could be worked with the students by means of gesture interaction with customized picture boards by teachers.

Another method cited was the method of Boquinhas phonovisuoarticulatory, developed by [29] with the aim of literacy and to rehabilitate reading and writing disorders. This method is based on the assumption that sensory integration of stimuli it is beneficial for the consolidation of learning. The PGCA would be used to interact with a communication board with photos of mouths by means of gestural interaction, as shown in Figure 5 B.

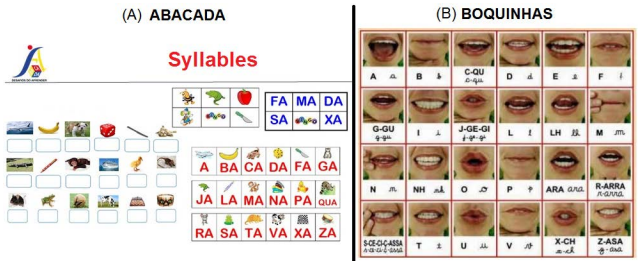


Figure 5: Methods ABACADA [53] and "Boquinhas" [43].

All the eight teachers considered the proposal interesting, valid, and even with doubts about its effectiveness, or application, did not measure efforts to assist in the information transfer. The expectation of one of the teachers is that our study could provide alternatives for his communications with students who do not carry out any voluntary movement or do not demonstrate any expression appropriate to the interactions that are made with them. Unfortunately, the current stage of our study do not aim at this public as our solution assumes the user has any movement that can be recognized. The system developed aims both to benefit from the knowledge of caregivers and to contribute with their communication strategies, being able to promote the effective participation of students in the classroom and their communication with teachers and other people.

5 CREATING CUSTOM DATASETS BY USERS WITH MOTOR AND SPEECH IMPAIRMENTS

After the first author had met several students with special needs registered in the four visited schools, a group of seven students were invited to participate in the experiment to create datasets with the system. The group was invited based on the interview results and recommendations from the teachers that suggested students who would have greater comprehension capacity and conditions to carry out voluntary movements. The invitation was sent to the students' families and one student was not authorized to participate. All invited students are characterized as people with cerebral palsy, with different levels of special needs as presented in Table 2.

Table 2: Characteristics of students with special needs selected to participate in the experiment

Stud.	Sex*	Age	Medical report
A	M	18 years old	Cerebral palsy due to sequels during labor.
B	F	29 years old	Brain damage and discrete hydrocephalus.
C	M	38 years old	Quadriplegia with athetosis component, bilateral sensorineural hearing loss.
D	F	20 years old	Pseudobulbar palsy. Generalized hypotonia and hyperreflexia.
E	F	18 years old	Static encephalopathy and spastic quadriplegia.
F	M	18 years old	Static encephalitis, epilepsy, and Taybi's Rubinstein Syndrome.

*Sex: M - Male; F- Female.

Each student who participated in the experiment was accompanied by a teacher who played the role of the "caregiver user" in the system, informing the gestures that the student usually uses to communicate and the meaning of each gesture. Therefore, tasks conducted by the teachers allowed us to evaluate the system in the creation of a dataset with gestures personalized for each student. The tasks expected to be performed during the experiment were: 1. creating the dataset by capturing gestures for training the system; 2. training and evaluating the system; 3. using the system to recognize gestures; 4. using gestures to select images in the communication board; and 5. using gesture for interacting with text editor or Internet browser.

For the experiment test with the PGCA system, the CNN-based classifier was used to gesture recognition, however, for each gesture, the video of the movement was stored, and later, the gesture recognition was simulated using also the SVM-based classifier. All actions performed in the system through gestural interaction were stored in a text file (log) in order to record events in the interaction with the interface. Images pertaining to students' datasets are presented only in the form of OF-MHI to preserve the participants' identity.

Student A. The Student A uses only two head gestures in the school environment to communicate with his/her classmates and teachers, which refer to "Yes" and "No". This student has a very preserved comprehension capacity. Data collection for the system training and the interaction tests with the interface were performed at two different sessions. For this student, the system was configured to use the "Zoom in" option in order to better capture facial movements. Student A presented some involuntary movements during the interaction with the system, leaning his leg or arm on the



Figure 6: Examples of gesture samples of Students A and B.

table where the computer and the camera were arranged, generating some samples with information about the back of the room, considered as noises. The dataset created with this student was composed of two classes; fifteen samples each class were considered valid to train the system. After training, the first step of interaction test with the interface was referring to the recognition of the gestures for which the system has been trained: the two gestures were correctly identified by the system when performed voluntarily by the student. Subsequently, the system was configured by associating the "Yes" class with the system confirm option. Such a configuration allowed us to test the communication board for writing words related to specific requests, such as "Sleep", "Bath", "Food", and others. In the following, the system was configured by associating the "No" class with the ENTER key, and the option "simulate keyboard" was selected, allowing the use of other applications. The configuration allowed to perform the selection of images on a communication board for writing words directly in a text editor, simulating the pressing of the keyboard's ENTER key when performing the "No" gesture. Some false positives were observed in this test, and to minimize this situation, a filtering by confidence level greater than 0.5 (certainty of a sample belonging to a particular class) was included to CNN-based classifier. The SVM-based classifier employs a confidence level of at least 0.5 by default. Figure 6 (A) presents examples of samples generated by Student A to compose his dataset.

Student B. Student B does not present significant motor impairment and uses hand and head gestures in the school environment, however, she avoids interacting with unknown people. She has some difficulties in the understanding capacity and is very shy, demanding constant encouragement from the teacher who accompanied her when performing the gestures. Data collection for system training was performed in a session in one day, and interaction tests with the interface were performed in another session three days later. The "Zoom in" option was disabled in the system. Her customized dataset was composed of five classes with thirteen samples each as shown in Figure 6 (B). After training, in the interaction test, the gestures referring to "Yes", "Fast" and "Bunny" (because of the Easter Bunny) were correctly recognized. The gestures referring to "Food" and "No" generated some erroneous interpretations, but this did not prevent the interaction with the system. Subsequently, the system was configured by associating the "Yes" class with the system confirm option, and the student selected images on a communication board composed of images related to handicrafts as it is a subject of interest of the student, according to her teacher. This board was used to write words associated with each figure in the system interface when the student performed the "Yes" gesture. Also, a communication board composed of images of vowels was used for the student to indicate the first letter in the "Elephant" word. With the help of the teacher, the student made the "Yes" gesture to select the figure referring to the letter "E" at the system interface and the corresponding sound was emitted. Figure 8 presents representations of tasks performed by the students A and B. Because the student demanded a lot of intervention from the teacher and had little initiative in selecting pictures on her own, no other interaction test was performed.

Student C. Student C presents motor and speech impairment and severe hearing loss, using hand and head gestures in the school environment. He uses some gestures of LIBRAS and home signs, but because of motor impairment in the hands, not all LIBRAS interpreters can understand his intentions of communication. Student C has his capacity for understanding well preserved and for the execution of the experiment he was accompanied by his LIBRAS interpreter and caregiver. Data collection for system training and interaction tests were performed on two different sessions. The "Zoom in" option was disabled. Initially, the dataset created by this student was composed of ten classes, with twelve samples each. Figure 7 shows examples of samples generated. After training, during the interaction test, some gestures characterized with similar movements were being confused by the classifier, such as the gestures "Yes", "I", "Mom" and "Water". The stored videos and samples related to these gestures showed the movements for these gestures are very similar to each other, varying only the position of the fingers or hand and, probably because of this, the training performed was not enough for the system to recognize them. Therefore, in another session one month later, a second dataset was created containing only seven gestures with seventeen samples each, leaving out the gestures "I", "Mon" and " Water ", and keeping the" Yes "gesture. With the new dataset, most of the gestures were correctly recognized by the system, except the gesture referring to "Bathroom"

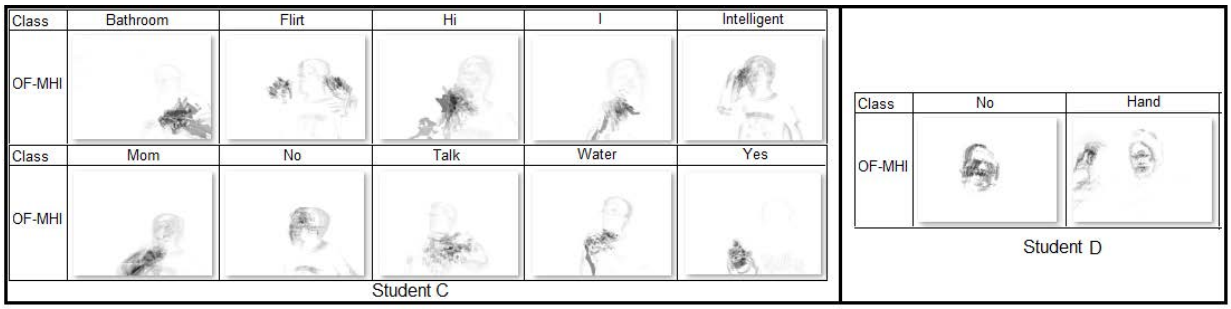


Figure 7: Examples of gesture samples of Students C and D.

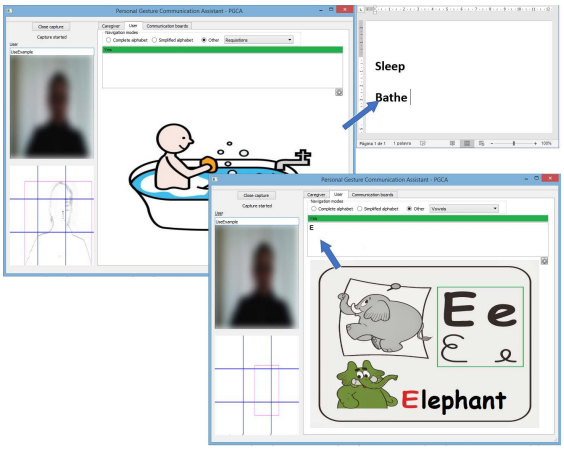


Figure 8: Communication boards used for students to select images [45] and write in a text editor and to indicate the initial letter of a word (Character of [55] adapted from [44])

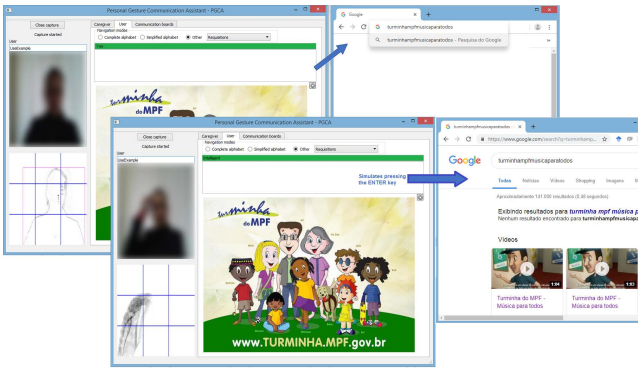


Figure 9: Communication board used to search on the web.

that was not recognized in some situations. Subsequently, the system was configured by associating the "Yes" class with the system confirm option, allowing the student to test the image selection on the communication boards. After, the system was configured by associating the "Intelligent" class with the ENTER key, and the "simulate keyboard" option was selected. Then, the student selected images on a communication board to write words in a text editor, and simulate the use of the ENTER key. It was also possible to select a figure on a communication board with keyword options to search in an Internet browser, writing directly in the browser URL and simulate the ENTER key to search for the keyword (Figure 9). The interaction test was completed in this step. However, following the same interaction structure, other gestures could be associated with the TAB key to navigate between the search results and select the desired page using the ENTER key simulation.

Student D. This student uses head gestures and facial expressions in the school environment, has a very preserved comprehension capacity and can voluntarily move the right arm despite many spastic movements. The gesture referring to "Yes" is a raise of eyebrows, however, as it has a lot of associated head movement, the system was not able to register correctly the movement of this facial expression. To create a dataset for this student to interact with the system, we chose to capture the movement referring to "No" (moving the head to both sides), and the movement referring to "Hand" (moving his right arm). Data collection was performed on a session, and interaction tests on another session five days later. Ten samples were considered valid for each class in dataset. After training, the two gestures were correctly identified by the system in the interaction test. Subsequently, the system was configured by associating the "Hand" class with the system confirm option and the "No" class with the ENTER key. The same interaction tests with the Internet Browser and Text Editor performed by Students A and C were performed by this student. However, some involuntary movements occurred with the arm used to make selections, and unintentionally the system selected items on the boards several times. According to the accompanying teacher, the use of the arm to give affirmative answers require more training.

Student E. The Student E uses only small head gestures and facial expressions to expose her communication intentions

in the school environment. The teachers expressed doubts about the student's level of understanding. Two unsuccessful attempts were made to collect data with this student in two different sessions, on different days. The student makes very restricted head movements and for the same intention to communicate "Yes", sometimes she moved her head and sometimes she just smiled. According to the teachers, when eating, the student puts her tongue out to indicate she does not want some specific food. But during the experiment, this gesture referring to "No" was never performed by the student, even after several attempts by the teacher asking questions seeking a negative answer. Therefore, the experiment with Student E was finished.

Student F. The Student F has progressively lost motor functions and uses hand and head gestures in the school environment, mainly pointing for objects of interest. According to the teachers, the student has a well-preserved comprehension capacity. The researcher who conducted the experiment participated in some classes with the student, observing the gestural interaction used by him. However, it was not possible to create a dataset for this student, even though two attempts were made to collect data on different dates and with the support of different teachers. In both sessions, despite having motor conditions the student showed no interest in carrying out the gestures when requested. After the second attempt to capture the gestures, the researcher questioned the student if he did not want to be filmed, and the student emitted a sound (considering his speech limitations) which was understood as a "No". Then, the data collection session was terminated.

Finally, datasets with personalized gestures performed by four students were created, and to evaluate classifiers' performance using these datasets, the k-fold Cross-Validation method was applied considering 10 folds, separating 90% of data for training and 10% for test. The number of training data was expanded by Data Augmentation where additional samples were created from existing data, creating others eight variations by rotating and scaling the original image. After running tests on all folds, it was calculated overall accuracy and Cohen's Kappa as shown in Table 3. The number of samples used to compose the student datasets varied according to the number of gestures captured and considered valid by the first author who carried out the experiment.

Table 3: Performance of classifiers: Accuracy, Cohen's Kappa, Standard deviation, Variance and False positives.

Classifier	Method	Student dataset				
		A	B	C1	C2	D
SVM-based (Confidence level >= 0.5)	Accuracy	1	0.86	0.87	0.87	1
	Cohen k	1	0.83	0.86	0.85	1
	Std. Dev.	0	0.09	0.13	0.08	0
	Variance	0	0.009	0.019	0.007	0
	False pos.	0	0.003	0.001	0.002	0
CNN-based (default)	Accuracy	0.97	0.78	0.73	0.87	1
	Cohen k	0.94	0.73	0.70	0.84	1
	Std. Dev.	0.030	0.140	0.107	0.119	0
	Variance	0.001	0.021	0.012	0.015	0
	False pos.	0.002	0.015	0.034	0.016	0
CNN-based (Confidence level >= 0.5)	Accuracy	0.97	0.73	0.68	0.82	1
	Cohen k	0.94	0.67	0.67	0.79	1
	Std. Dev.	0.030	0.200	0.121	0.147	0
	Variance	0.001	0.044	0.016	0.024	0
	False pos.	0.002	0.012	0.012	0.011	0

6 DISCUSSION AND LEARNED LESSONS

The number of students who participated in the experiment was lower than expected because in first contact with the schools, several students were presented as possible candidates (i.e., people with motor and speech impairments). However, as the researcher became better acquainted with these students, it was found out that the minority of the students with these characteristics does not have associated cognitive impairment. Even in situations where there are similar diagnoses (e.g. people with cerebral palsy), teachers themselves have doubts about the level of understanding each student effectively has. Then, many students initially considered as potential participants of the experiment were not selected, because it was imperative the user to have a good understanding both to accept taking part in the experiment and to benefit from the intentional use of the system.

By observing the routine of students who do not speak in the school environment and the information obtained through the interviews, it was possible to perceive these students end up having a very passive posture because their gestures can only be interpreted if the teacher is looking at them. In the case of students who performs only "Yes" and "No" movements, it is only possible for that student to interact with the teacher or classmates when they make him/her affirmative or negative questions. Therefore, the potential for the proposed system in allowing students to interact via personalized gestures with communication boards is evident. Such interaction possibilities allow students to express their intentions of communication, drawing the attention of the teacher when necessary (using voice synthesizer), gaining some level of independence and power of interaction in the environment in which they live.

The observations and results from the experiments evinced aspects to be improved in the developed system. The interface can be improved by adding the most visible signaling (compared to a progress bar currently used) to indicate the user whether the system considered the gesture as completed or is trying to recognize it. Messages textually presented by the interface could be translated to LIBRAS and be presented by means of synthesized voice in order to facilitate the understanding by deaf users and by non-literate users. For the students who have the capacity to execute a restricted number of voluntary gestures, it is necessary greater attention of the caregiver in order to initiate and finalize the capture

process through the interface. In the tests with Student A, when he was observing the classmates in the class, moving the head naturally without the intention of communicating something, the system ended up identifying the movement as "Yes" or "No" generating unnecessary and incorrect answers. This was because the system remained with the capture process started even after the teacher ended the interaction with it. Since the student did not have other gestures that could be associated with the capture/start functionality, it must be done by the caregiver.

In the case of Student C, the distinct amount of signals he is able to make is quite large, however, some of the signs generated very similar motion representations that ended up confusing the classifier. For gesture-based interfaces, the precision (high true positives and low false positive rates) has to be assured while maintaining the natural feeling of interpersonal communication [33]. To minimize false positives, the PGCA system uses a confidence level that can be customized to avoid false assignments. Moreover, users can create datasets composed of gestures that are quite similar to each other and, in these cases, if the system can not recognize these gestures correctly, the system may suggest a new data collection for these gestures or suggest only one of the similar gestures to compose the dataset. Further studies are being conducted to improve overall system accuracy by including more information on the motion representation, in order to enrich the input samples used by the classifier.

Because different sessions occurred on different days, some students performed gestures with different speed and intensity, suggesting it is interesting to collect samples on different days to generate more information to be learned by the classifier. In addition, considering the situation observed in the attempt to capture samples of Student F, making the system interface more attractive, perhaps using a serious game that requires the execution of similar movements for its use, can be a way to stimulate the interest of students and promoting engagement and motivation to train and use the system.

Although the calibration process was a concern because it requires specific positioning of the users in front of the camera, no problem was found. All students had motor conditions to move their head and when the user's nose was positioned near the center of the screen (even on the move), the requirement of the calibration step was easily achieved. The inclusion of facial landmarks as points to be tracking by Optical Flow was made after the interview with some teachers, who reported that selected students could perform very discreet facial movements, such as raising their eyebrows, blinking their eyes or showing their tongue. This situation was not predicted in the first version of the system, but the change implemented made it possible to track these small movements satisfactorily. Transfer Learning has proven to be efficient for our study because even a trained network with a dataset composed of colored images can be customized to successfully recognize our grayscale images. However, the SVM-based classifier presented a slightly better performance on gesture recognition of the student datasets.

The experiment allowed to evaluate the technical feasibility of the proposal and, in the future, after the system improvements and a training phase, new experiments will be carried out with the target audience to evaluate the effectiveness of the use of the system in daily life of people with AAC needs. After the experiments, the classifier that shows the best overall results is the natural candidate to be adopted for the software final version.

The PGCA system in its current version may not yield satisfactory results for the recognition of complex signal languages, composed of a few hundred distinct gestures with few variations between them. However, the methodology that was used as a basis for the development of the system could be used to develop other approaches, taking into account different training and recognition strategies, and the characteristics of the gestures employed as a means of communication by the intended audience.

7 CONCLUSION

In this paper, it was presented the PGCA: a Computer Vision system developed as assistive technology to support AAC based on personalized gestural interaction. The paper presents and describes the system, and discusses the results of interviews with professionals of the special education field and from an experiment carried out with students with motor and speech impairment. Five customized datasets were generated by students and their caregivers. Two classifiers were evaluated to gesture recognition: SVM-based classifier and CNN-based classifier. The SVM-based classifier presented a slightly better performance to recognize gestures from the generated datasets.

Results presented by the classifiers and the interaction tests performed by students with the PGCA system suggest the system is promising to support AAC. As for directions for future work, the system will be improved and after a training session to familiarize users with the system, new tests will be conducted with the target audience and the effectiveness of the final system as assistive technology will be evaluated.

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